Some new faces of Bregman divergences

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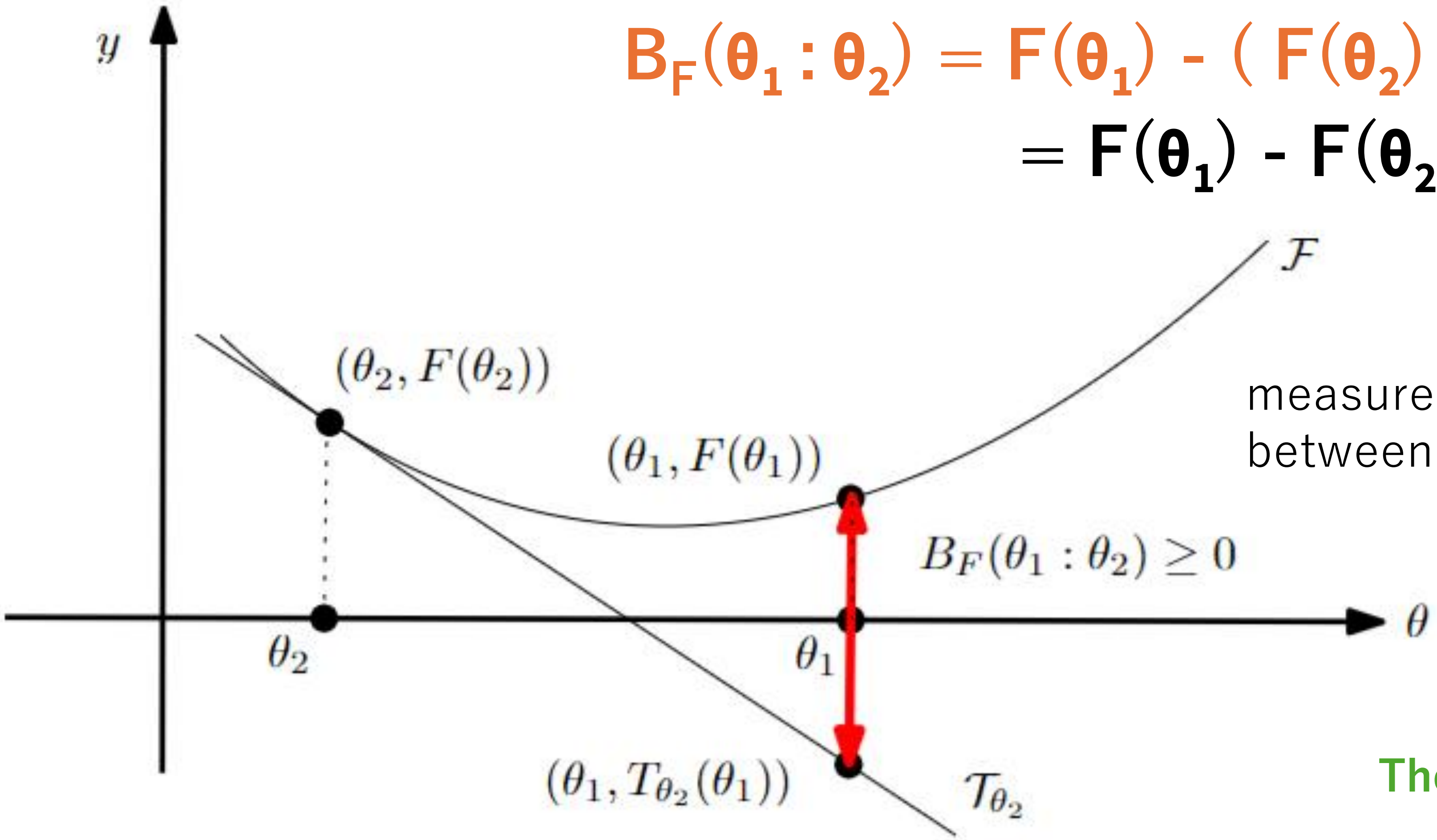
Bregman divergences (1960's)

- $F: \Theta \subseteq \mathbb{R}^m \rightarrow \mathbb{R}$ a strictly convex and smooth real-valued function on a finite-dimensional Hilbert space equipped with inner product <.,.>

Bregman divergence $B_F: \Theta \times \mathrm{Int}(\Theta) \rightarrow \mathbb{R}$

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - (\, F(\theta_2) + \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle \,)$$

$$= F(\theta_1) - F(\theta_2) - \langle \theta_1 - \theta_2, \nabla F(\theta_2) \rangle$$



measures a vertical convexity gap between graph of F and one of its tangent

The Bregman chord divergence, GSI'19

Bregman divergences unify various distortions

$$B_F(\theta_1 : \theta_2) = F(\theta_1) - F(\theta_2) - < \theta_1 - \theta_2 , \nabla F(\theta_2) >$$

- Quadratic negentropy generator yields ***squared Euclidean divergence:***

$F(\theta) = ½ \ \theta^T Q \ \theta$ with $\Theta = R^d$

$\nabla F(\theta) = Q\theta$

$B_F(\theta_1 : \theta_2) = ½ \ (\theta_2 - \theta_1)^T \ Q \ (\theta_2 - \theta_1) = ½ \ || \ \theta_2 - \theta_1 ||^2_Q$

Computational geometry L_2^2

- Shannon negentropy generator yields **relative entropy** aka. *Kullback-Leibler divergence :*

$F(\theta) = \Sigma_i \ \theta_i \log(\theta_i)$ with $\Theta = \Delta^d$ (standard/probability simplex)

$\nabla F(\theta) = [\log(\theta_i) - \theta_i]_i$

$B_F(\theta : \theta') = \Sigma_i \{ \ \theta_i \log(\theta_i / \ \theta_i') + \theta_i' - \theta_i \ \} = \Sigma_i \ \theta_i \log(\theta_i / \ \theta_i')$

$= KL(\theta_i : \theta')$ since $\Sigma_i \ \theta_i = \Sigma_i \ \theta_i' = 1$

Bring CG in Information geometry!

New faces of Bregman divergences

- BDs viewed as **mixed-parameterization power distances**

→ Revisit KL minimax problems using Laguerre geometry

- BDs and quadratic polarities

→ BDs as inner products using two extra dimensions (gauge freedom)

- Functional Bregman divergences, **kernelized functional BDs**

→ Generalize maximum mean discrepancy, 2-sample hypothesis testing

Convex duality: Mixed parameterizations of BDs

- Legendre-Fenchel transform of a convex function F:

$$F^*(\eta) = \sup_{\theta \in \Theta}\{<\theta, \eta> - F(\theta)\}$$

- Young inequality: $F(\theta_1) + F^*(\eta_2) \geq <\theta_1, \eta_2>$

  with equality iff. $\eta_2 = \nabla F(\theta_1)$

- Build the Fenchel-Young divergence from this inequality: lhs-rhs ≥0

$$Y_{F,F^*}(\theta_1, \eta_2) = F(\theta_1) + F^*(\eta_2) - <\theta_1, \eta_2> \geq 0$$

- Mixed and dual parameterizations of an underlying geometric divergence $\eta = \nabla F(\theta)$ and $\theta = \nabla F^*(\eta)$

$$B_F(\theta_1 : \theta_2) = Y_{F,F^*}(\theta_1 : \eta_2) = Y_{F^*,F}(\eta_2, \theta_1) = B_{F^*}(\eta_2 : \eta_1)$$

Bregman divergences as power distances

Start with FY divergence $Y_F(\theta:\eta') := F(\theta)+F^*(\eta')-\langle\theta,\eta'\rangle$

and use identity $-\langle\theta,\eta'\rangle = \frac{1}{2}\left(\|\theta-\eta'\|^2-\|\theta\|^2-\|\eta\|^2\right)$

to get $B_F(\theta:\theta') = Y_F(\theta:\eta') = \frac{1}{2}\sigma(\hat{\theta}:\check{\eta}')$

Power distance $\sigma(\hat{p},\hat{p}') := \|p-p'\|^2-w-w'$

(signed measure)

$\hat{p}=(p,w) \quad \hat{p}'=(p',w')$

Weighted points expressed in **mixed parameterizations**:

$\hat{\theta}=(\theta,\|\theta\|^2-2\,F(\theta)) \qquad \check{\eta}=(\eta,\|\eta\|^2-2\,F^*(\eta))$

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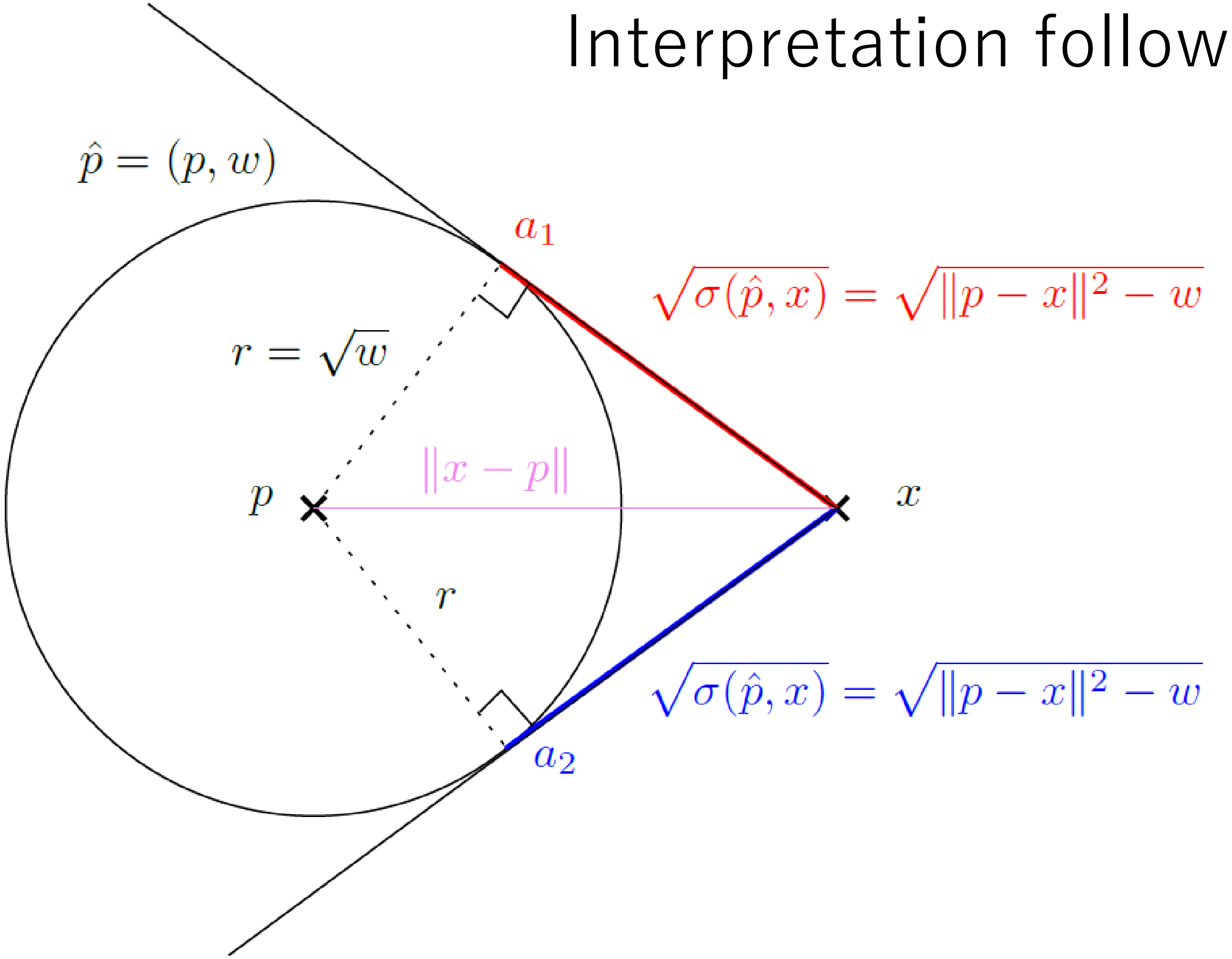
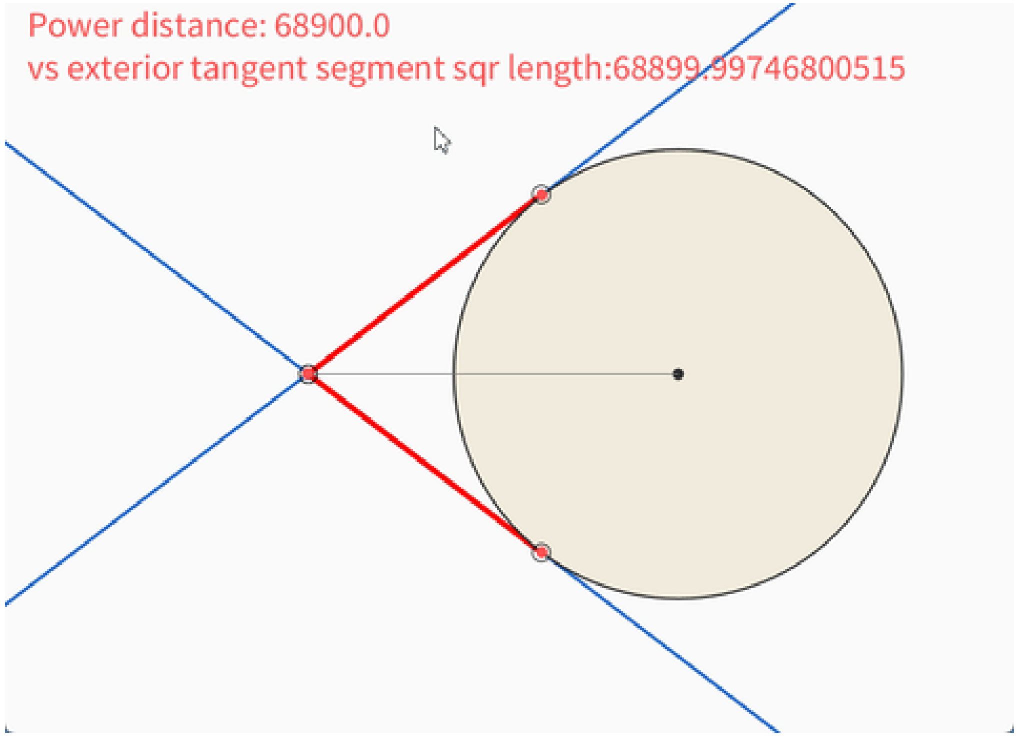
Visualizing power distances $\sigma(\hat{p}, \hat{p}') := \|p - p'\|^2 - w - w'$

(when one point x has zero weight)

Interpretation follows from the Pythagoras' theorem





Laguerre geometry

Power distance is negative when x inside the disk

KL minimax: Redundancy, a fundamental quantity

$$R(\mathcal{P}) := \min_{Q\in\mathcal{Q}} \max_{P\in\mathcal{P}} D_{\mathrm{KL}}(P:Q)$$

"Minimize maximum error"

Radius of KL miniball

- universal compression (source coding)
- online learning (sequential prediction)
- hypothesis testing
- minimum description length principle
- gambling
- model integration/fusion

$$\mathcal{P} = \{P_1, \ldots, P_n\} \subset M_1^+(\mathcal{X})$$

$$\mathcal{Q} = M_1^+(\mathcal{X})$$

$$D_{\mathrm{KL}}(P:Q) = E_P[\log \frac{1}{Q(X)}] - H(P)$$

Swap rhs opt. to left opt.

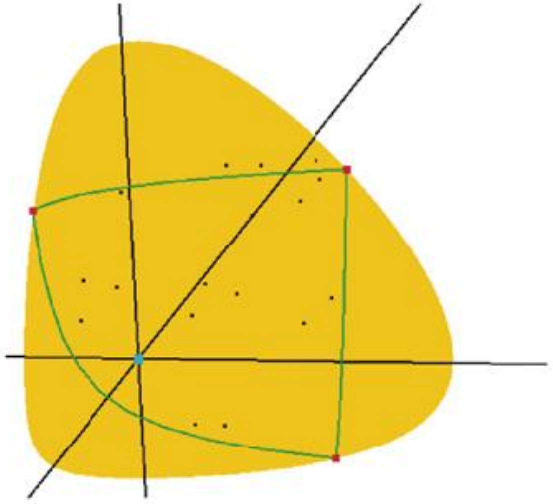
$$\mathcal{E} = \{P_\theta : \theta \in \Theta\} \qquad D_{\mathrm{KL}}(P_{\theta_i} : P_\theta) = B_F(\theta : \theta_i)$$

When distributions belong to a same exponential family with cumulant function F

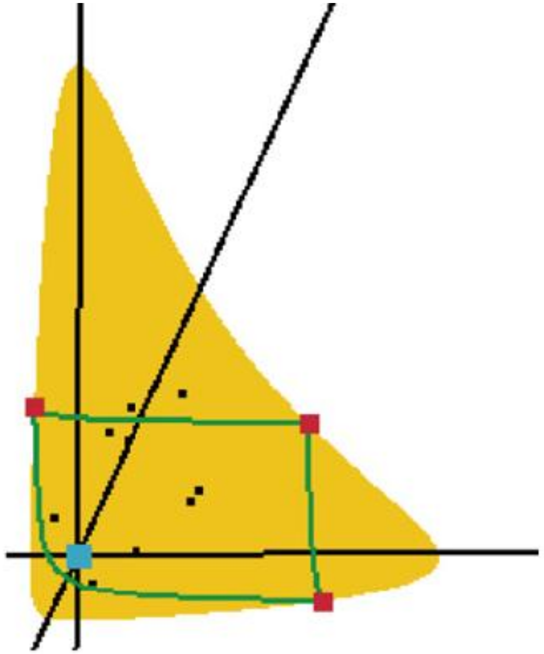
$$R(\mathcal{P}) := \min_{Q\in\mathcal{Q}} \max_{P\in\mathcal{P}} D_{\mathrm{KL}}(P:Q) \Rightarrow \min_{\theta\in\Theta} \max_{i\in[n]} B_F(\theta:\theta_i)$$

Smallest/Minimum enclosing Bregman balls

$$\min_{\theta \in \Theta} \max_{i \in [n]} B_F(\theta : \theta_i)$$

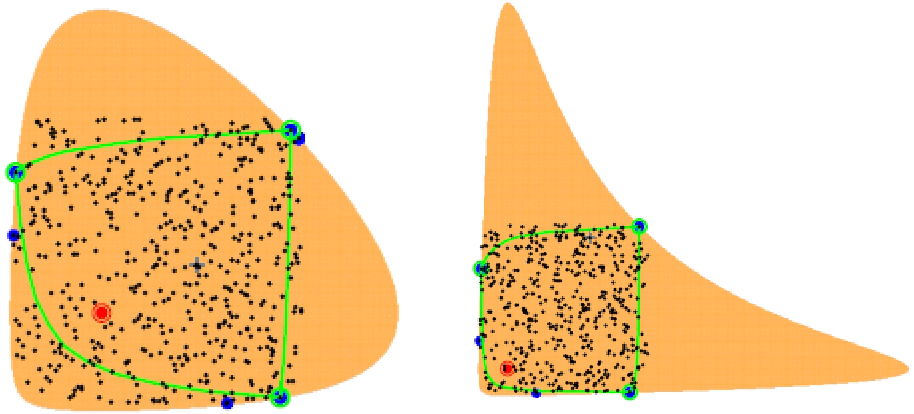


Kullback-Leibler



Itakura-Saito

LP-type



Algorithm 2: BBC($\mathcal{S}$)

Input: Data $\mathcal{S} = \{\mathbf{s}_1, \mathbf{s}_2, ..., \mathbf{s}_m\}$;

Output: Center $\mathbf{c}$;

Choose at random $\mathbf{c} \in \mathcal{S}$;

for $t = 1, 2, ..., T-1$ **do**

- $\mathbf{s} \leftarrow \arg\max_{\mathbf{s}' \in \mathcal{S}} D_F(\mathbf{c}, \mathbf{s}')$;
- $\mathbf{c} \leftarrow \boldsymbol{\nabla}_F^{-1}\left(\frac{t}{t+1}\boldsymbol{\nabla}_F(\mathbf{c}) + \frac{1}{t+1}\boldsymbol{\nabla}_F(\mathbf{s})\right)$;

Frank-Wolfe type

"On the smallest enclosing information disk." Information Processing Letters 105.3 (2008)
"Fitting the smallest enclosing Bregman ball." ECML 2005.

Bregman MEB circumcenter as a power MEB circumcenter

Bregman circumcenter

Power circumcenter

$$\boxed{\theta^* = p^*}$$

$$\begin{aligned}\theta^* &= \mathrm{argmin}_{\theta\in\Theta} \max_{i\in[n]} B_F(\theta : \theta_i)\\ &= \mathrm{argmin}_{\theta} \max_{i} Y_F(\theta : \eta_i)\\ &= \mathrm{argmin}_{\theta} \max_{i} \frac{1}{2}\sigma(\hat{\theta} : \check{\eta}_i)\\ &= \mathrm{argmin}_{p} \max_{i} \sigma(p : \hat{p}_i) := p^*\end{aligned}$$

Weighted points:

$$\hat{p}_i = \check{\eta}_i$$

$$\check{\eta}_i = (\eta_i = \nabla F(\theta_i), \|\eta_i\|^2 - 2\,F^*(\eta_i))$$

Power distance

$$\sigma(p, \hat{p}') := \|p - p'\|^2 - w'$$

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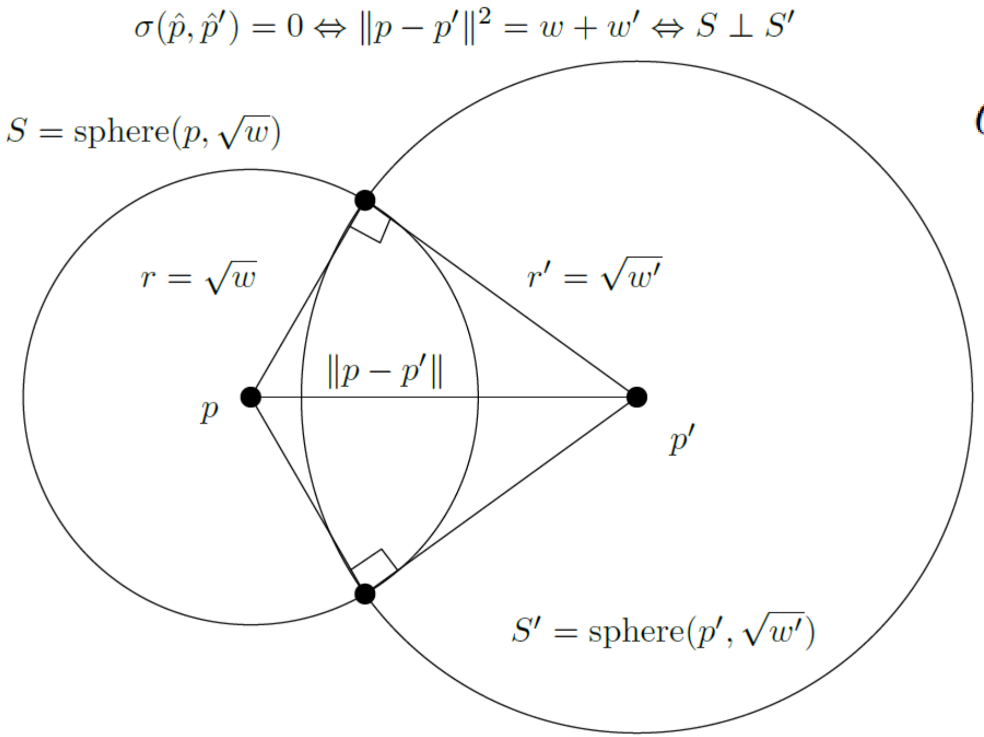
Vanishing power distance = Sphere orthogonality



$$\sigma(\hat{p},\hat{p}') := \|p-p'\|^2 - w - w'$$

Pythagoras' theorem:

Power distance vanishes when spheres are orthogonal

Power circumcenter and support points

$$\sigma(\theta^*, \hat{p}_i) = w^* \Rightarrow \sigma(p^*, \hat{p}_i) = 0$$

$$p^* = (\theta^*, w^*)$$

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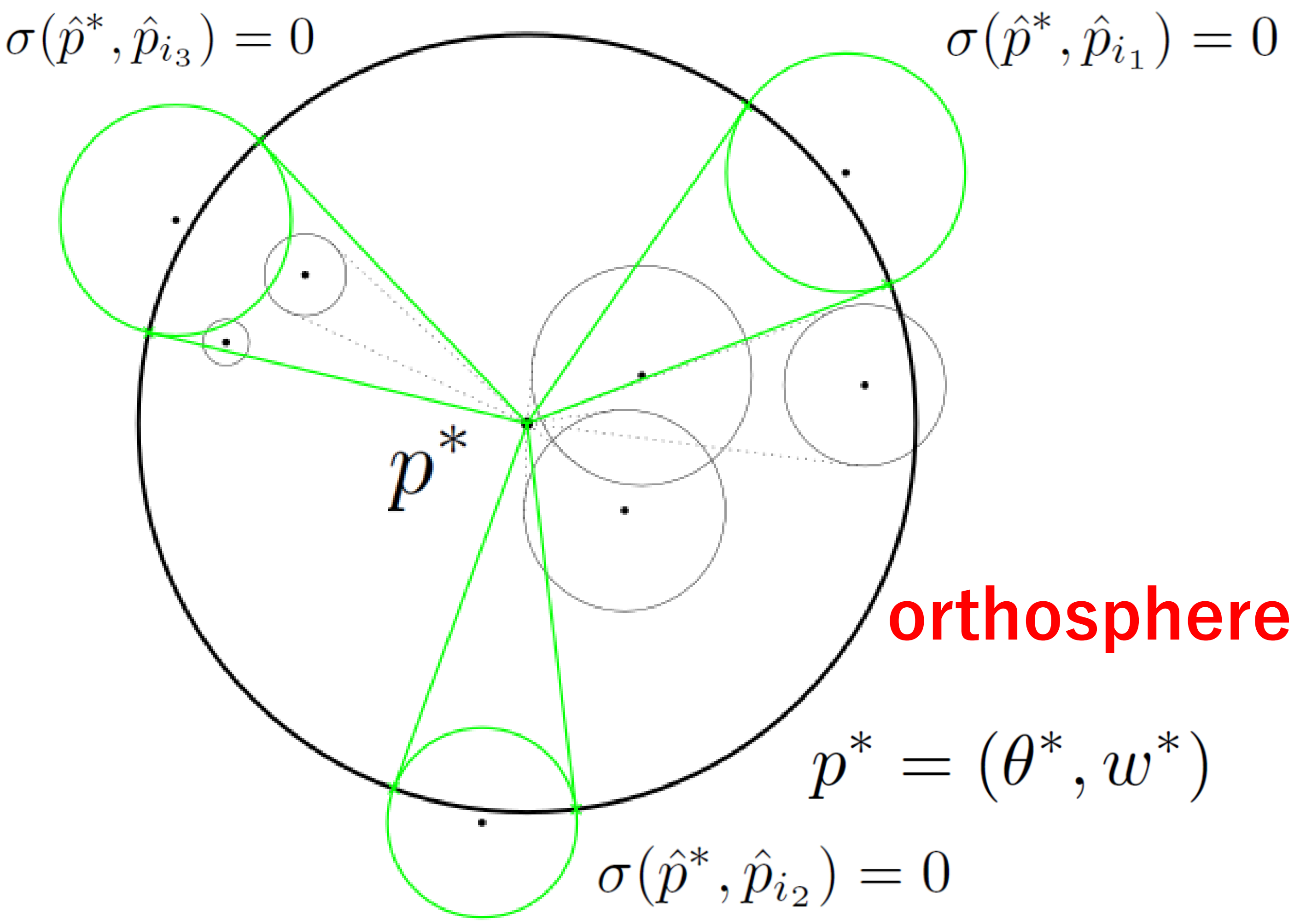
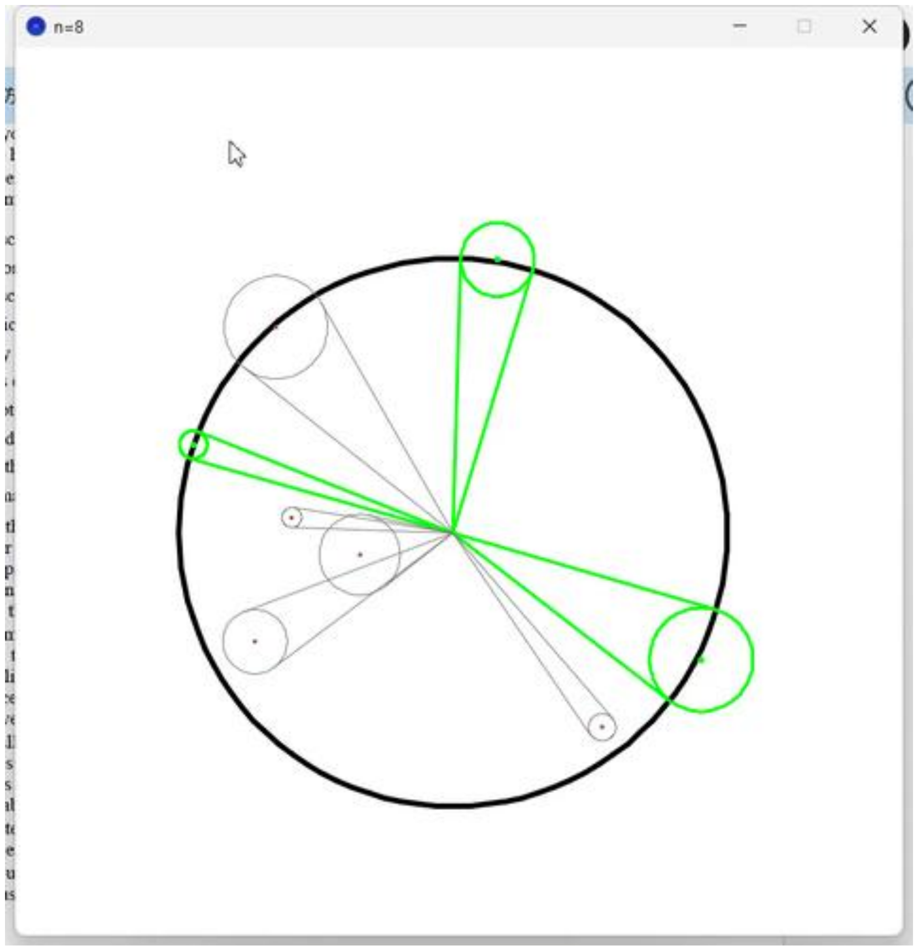
Bregman MEBs as power MEBs

Use computational geometry toolbox in information geometry!





Bregman miniball in θ as Power miniball in η

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Frank-Wolfe coreset power MEB approximations

Algorithm 4: Left Bregman MEB approximation algorithm BregmanBC($\mathcal{T} = \{\theta_i\}$,$T$)

Choose at random $c_1 \in \mathcal{T}$;
for $t = 1, 2, \ldots, T-1$ **do**
  $f_t \leftarrow \arg\max_{i\in[n]} B_F(c_t : \theta_i)$;
  $c_{t+1} \leftarrow (\nabla F)^{-1}\left(\frac{t}{t+1}\nabla F(c_t) + \frac{1}{t+1}\nabla F(\theta_{f_t})\right)$;
end
Return $\tilde{c} = c_T$;

θ -view

Gradient map $\eta = \nabla F(\theta)$

Algorithm 5: BregmanBCEta($\mathcal{E} = \{\eta_i\}$,T)

Choose at random $c_1 \in \mathcal{E}$;
for $t = 1, 2, \ldots, T-1$ **do**
  $f_t \leftarrow \arg\max_{i\in[n]} B_{F^*}(\eta_i : c_t)$;
  $c_{t+1} \leftarrow \frac{t}{t+1} c_t + \frac{1}{t+1}\eta_{f_t}$;
end
Return $\tilde{c} = c_T$;

η -view

Algorithm 6: FWPowerMEB($\hat{\mathcal{S}} = \{\hat{p}_i = (p_i, w_i)\}$, T)

Choose at random $c_1 \in \{\xi_i\}$;
for $t = 1, 2, \ldots, T-1$ **do**
  $f_t = \leftarrow \arg\max_{i\in[n]} \sigma(c_t, \hat{p}_i)$;
  $c_{t+1} = \frac{t}{t+1} c_t + \frac{1}{t+1} p_{f_t}$;
end
Return $\tilde{c} = c_T$;

Weighted η –point view

Power distance view

Facilitates the Frank-Wolfe analysis and variants

$$\arg\max_{i\in[n]} B_{F^*}(\eta_i : c_t) = \arg\max \sigma(c_t, \hat{q}_i) \qquad \hat{q}_i = (\eta_i, 2\omega_{F^*}(\eta_i) = \|\eta_i\|^2 - 2F^*(\eta_i))$$

Fenchel-Young divergences as inner products

F a Legendre-type strictly convex function with convex conjugate F*, (F*)*=F
Fenchel-Young divergences are equivalent to Bregman divergences:

$$
\begin{aligned}
Y_F(\theta_1 : \eta_2) &:= F(\theta_1) + F^*(\eta_2) - \langle \theta_1, \eta_2 \rangle = Y_{F*}(\eta_2 : \theta_1) \geqslant 0 \\
&= \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle
\end{aligned}
$$

GPU!

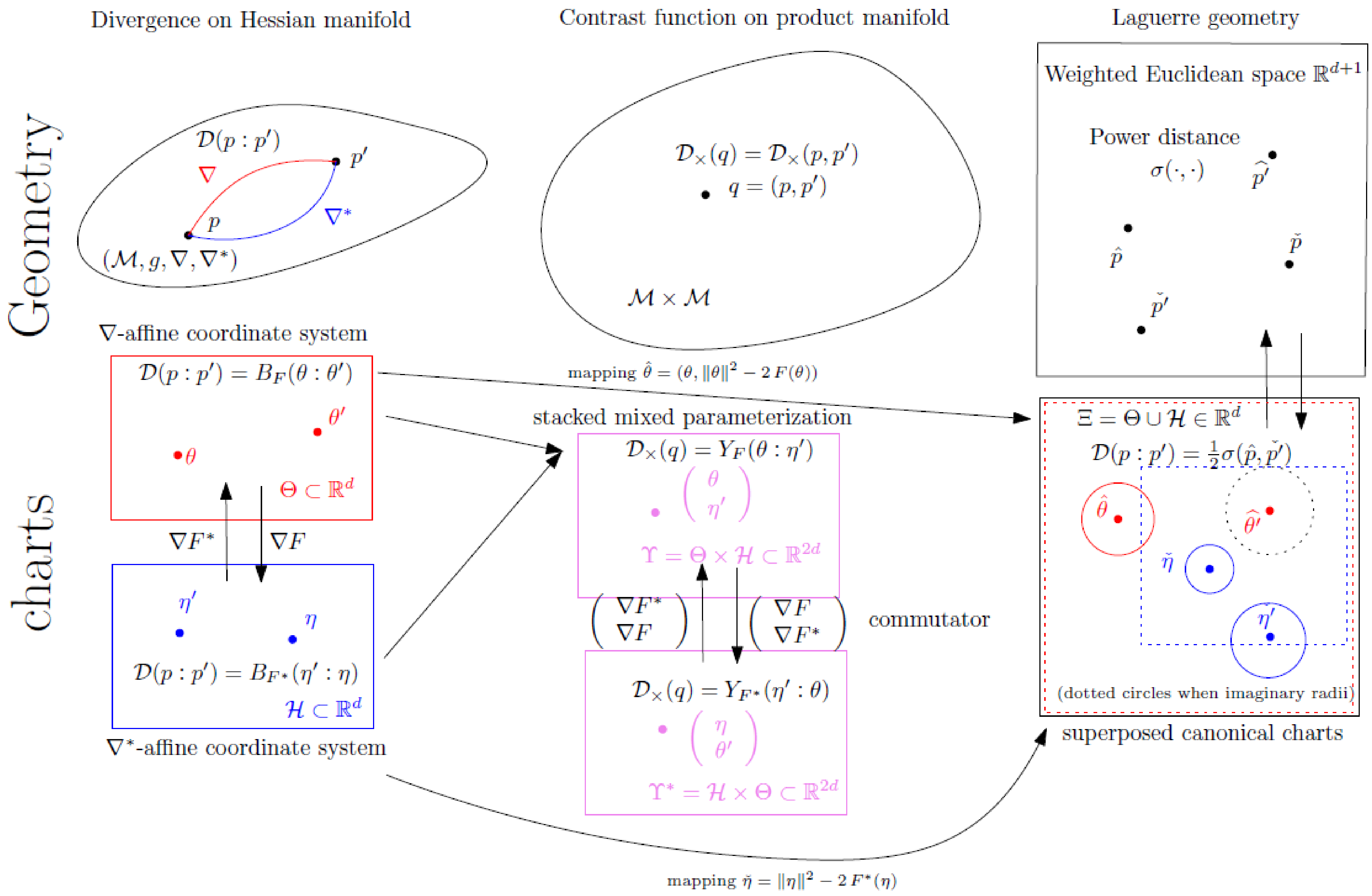
Geometric interpretation: Legendre transform considered as a polarity

$$
Y_F(\theta_1 : \eta_2) = (\tilde{\theta}_1^{\uparrow})^{\top} C_{\mathcal{L}} \tilde{\eta}_2^{\uparrow}, \quad C_{\mathcal{L}} := \begin{bmatrix} -I & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \Longrightarrow Y_F(\theta_1 : \eta_2) = \left\langle \begin{bmatrix} -\theta_1 \\ 1 \\ F(\theta_1) \end{bmatrix}, \begin{bmatrix} \eta_2 \\ F^*(\eta_2) \\ 1 \end{bmatrix} \right\rangle
$$

where $\tilde{\theta}^{\uparrow} := \begin{bmatrix} \theta \\ F(\theta) \\ 1 \end{bmatrix}, \quad \tilde{\eta}^{\uparrow} := \begin{bmatrix} \eta \\ F^*(\eta) \\ 1 \end{bmatrix}$ are homogeneous vectors of points on the graphs of potential functions

Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv: 2606.09077

Geometric interpretations



Laguerre geometry

Edmond Nicolas Laguerre

Born 9 April 1834
Bar-le-Duc, France

Died 14 August 1886 (aged 52)
Bar-le-Duc, France

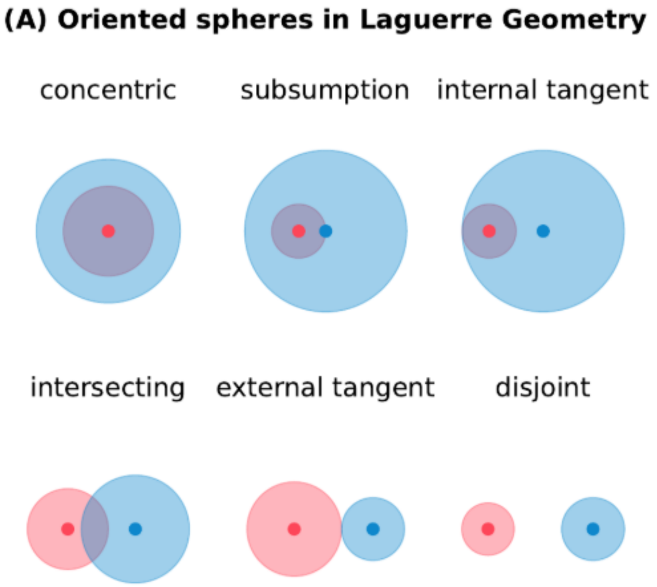
Laguerre geometry is a branch of mathematics founded by French mathematician Edmond Laguerre that studies the properties of **oriented hyperplanes and oriented spheres** under Laguerre transformations.

https://en.wikipedia.org/wiki/Laguerre_transformations

[Submitted on 12 Jul 2026]

Laguerre Geometry for Interpreting Large Language Models

Chunwei Ma, Russell Wolfinger



https://horsepurve.github.io/interpret-llm/posts/01_llm_lens/